



Advances in isothermal amplification: novel strategies inspired by biological processes



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ABSTRACT

Nucleic acid amplification is an essential process in biological systems. The *in vitro* adoption of this process has resulted in powerful techniques that underpin modern molecular biology. The most common tool is polymerase chain reaction (PCR). However, the requirement for a thermal cycler has somewhat limited applications of this classic nucleic acid amplification technique. Isothermal amplification, on the other hand, obviates the use of a thermal cycler because reactions occur at a single temperature. Isothermal amplification methods are diverse, but all have been developed from an understanding of natural nucleic acid amplification processes. Here we review current isothermal amplification methods as classified by their enzymatic mechanisms. We compare their advantages, disadvantages, efficiencies, and applications. Finally, we mention some new developments associated with this technology, and consider future possibilities in molecular engineering and recombinant technologies that may develop from an appreciation of the molecular biology of natural systems.

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1. Introduction

Nucleic acid amplification is an essential process in biological systems. The process enables production of genetic copies to be passed onto future generations for maintaining life. Nowadays, nucleic acid amplification is not simply a biological process for scientific investigation, but its *in vitro* adaptation has become a useful tool in research, clinical diagnosis, forensic science, agriculture, epidemiology and many other fields (Moore, 2005). Laboratory *in vitro* adaptations of nucleic acid amplification began with the advent of polymerase chain reaction (PCR), which relies on heating and cooling of samples. Recently, new methods that do not require heating and cooling, known as isothermal methods, have been reported.

Here we review the well-established isothermal amplification methods developed over the last two decades. While previous reviews of the technology have emphasised specific applications (Asiello and Baeumner, 2011; Chang et al., 2012; Morisset et al., 2008; Zanolli and Spoto, 2012), here we focus on the underlying enzymatic reaction mechanisms as adaptations of natural biological processes. This focus can guide molecular engineering approaches for further improved isothermal amplification processes and applications. The review begins with a discussion of non-isothermal nucleic acid amplification, namely PCR, and then discusses isothermal amplification methods inspired by RNA transcription and DNA replication. We then describe some novel methods, engineered from the ever-expanding toolkit of adapted

molecular processes, before considering future possibilities as our molecular engineering capabilities continue to increase.

2. Non-isothermal nucleic acid amplification

PCR is the best known nucleic acid amplification method, developed by Kary Mullis in 1983 (Mullis et al., 1986). This method can generate billions of DNA copies from a single target molecule within a few hours. The reaction requires no more than a test tube, a few simple reagents, and a heat source. PCR employs three key steps: initiation, annealing and elongation, all of which interplay with an increasing or decreasing of temperature. The process firstly raises the temperature to 95 °C to separate the double-stranded DNA, and secondly reduces the temperature to 50–65 °C, which allows the annealing of two primers (a forward and a reverse primer) (Mullis and Faloona, 1987). Thirdly, elevating the temperature to 70–74 °C enables the *Thermus aquaticus* (Taq) DNA polymerase to elongate the templates (Fig. 1) (Saiki et al., 1988). After running about thirty repeating cycles, the starting few copies of DNA have been replicated exponentially to approximately a billion copies. Such powerful augmentation hugely benefits sample preparation of nucleic acids for research, especially when the amount of DNA sample is marginal.

Nevertheless, in pursuing contemporary nucleic acid amplification towards point-of-care (POC) applications, PCR has some significant disadvantages. The main problem is that PCR is non-isothermal, which

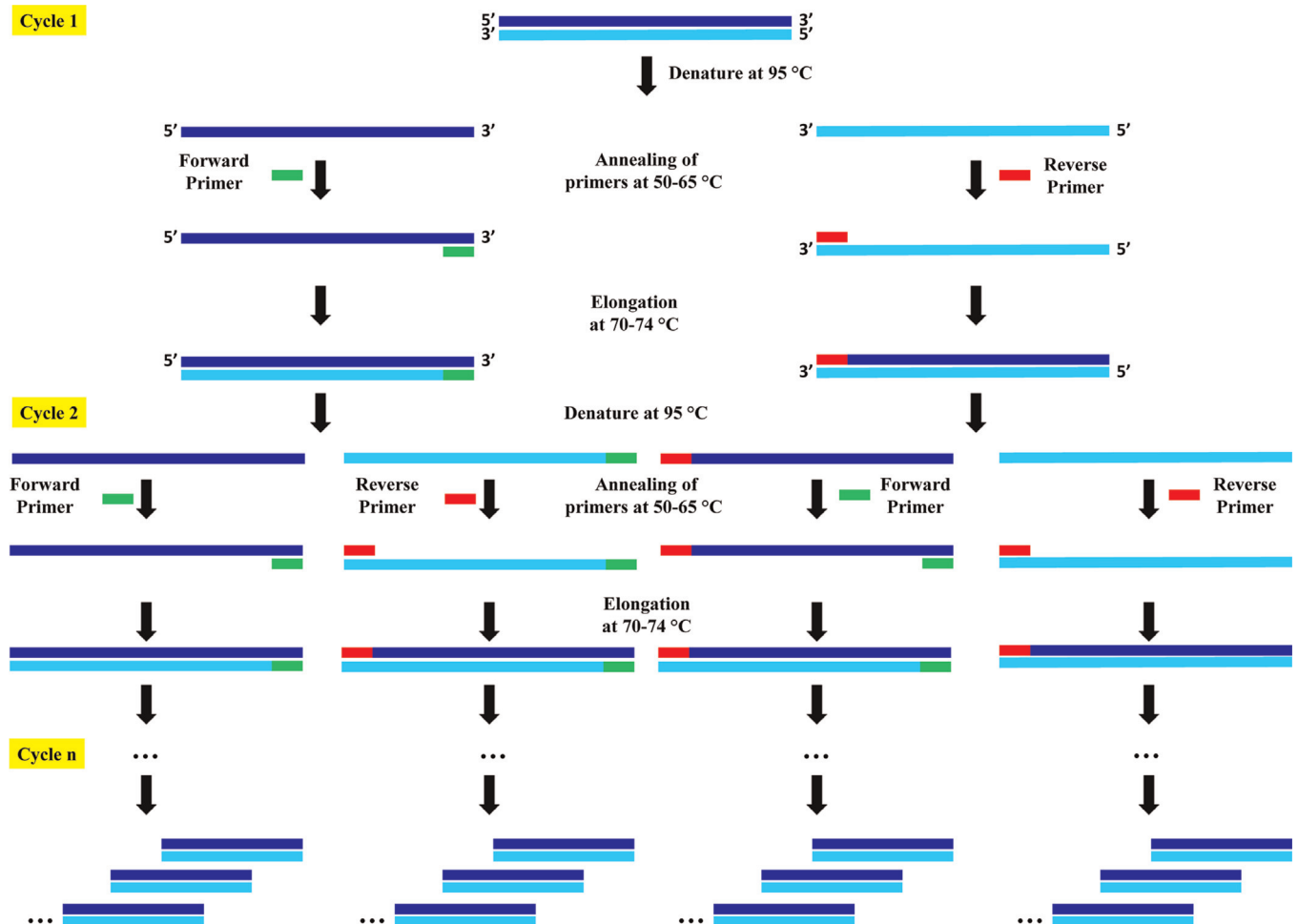


Fig. 1. Principle of PCR. The double-stranded DNA is denatured to separate single strands at 95 °C. The temperature is decreased to 50–65 °C to allow the two primers to anneal to the single strands. After that, the temperature is increased to 70–74 °C for the DNA polymerase to elongate the chains. Then the whole process recycles and the DNA replicates exponentially.

The continuous cycles from the RNA– result in exponential amplification of RNA– and cDNA products; approximately 10^6 – 10^9 -fold increases are obtained within 90 min. In contrast to the original 3SR system, the addition of dimethyl sulphoxide (DMSO) and the elevation of the reaction temperature from 37 °C to 41 °C in NASBA endow higher reaction specificity and prominent production of RNA molecules of desired length (Deiman et al., 2002). In addition to RNA amplification, NASBA can be used to amplify DNA analytes, but requires two denaturing steps. Firstly, the double-stranded DNA is denatured (at 95 °C) to allow P1 to bind and extension is completed with AMV reverse transcriptase. The second denaturing step enables P2 to bind to the extension product of P1. All three of these enzymes have to be added again afterwards, as they became inactivated during the second denaturation (Deiman et al., 2002).

Detection of NASBA amplicons is mainly end-point based. Similar to PCR, the amplicons can be classically separated by gel electrophoresis and are stained with ethidium bromide for visualisation under UV. A denaturant is usually required to eliminate the formation of secondary structures of RNA amplicons and the staining efficiency of ethidium bromide on RNA is not as effective as with DNA (Jean et al., 2001). Alternatively, since the NASBA amplicons are single-stranded RNAs, hybridisation with sequence-specific probes can be employed for detection, such as electrochemiluminescence (ECL) (van Gemen et al., 1994) and enzyme linked gel assay (ELGA) (Samuelson et al., 1998). The first method, ELC, involves two probes: a capture probe that is attached to magnetic beads to immobilise the amplicon, and a ruthenium labelled detection probe that hybridises with the amplicon at a distinct site other than the capture probe; the ELC signal is measured from voltage changes due to the redox reaction upon ruthenium (van Gemen et al., 1994). The latter method (ELGA) utilises horseradish peroxidase (HRP)-labelled probes: the free and hybridised probes are separated by electrophoresis in a gel containing dextran sulphate; incubation of the gel with a substrate for HRP gives rise to a coloured visualisation (Samuelson et al., 1998). To achieve faster end-point detection, NASBA is frequently coupled with a lateral flow device for detecting RNA viruses (e.g. HIV-1) by virtue of its amplification mechanism (Rohrman et al., 2012). A more elegant detection is by molecular beacon (Fig. 3) (Leone et al., 1998). The stem-loop structured probe contains a fluorophore at the 5' end and a quencher at the 3' end. The hairpin loop sequence is complementary to the target sequence of the

amplicon. At this state, no fluorescence is produced as the quencher absorbs all the energy from the fluorophore. Binding the probe to the amplicon opens up the hairpin loop, causing separation of the fluorophore from the quencher, hence light emitted from the fluorophore can be detected (Tyagi and Kramer, 1996). In essence, applying molecular beacons permits real-time detection, making simultaneous quantification and kinetic study plausible in NASBA.

Advantages of NASBA include: (1) the predominant production of RNA makes NASBA useful in RNA virus diagnosis; (2) circumventing the additional reverse transcription step required by PCR in amplifying RNAs, thus avoiding contamination and reducing hands-on time (Katherine Loens et al., 2005); (3) a low reaction temperature (41 °C) that requires a lower power supply; and (4) the amplicon yield is influenced less by primers compared to that of PCR, with the level of RNA amplicons exceeding the level of primers by at least one order of magnitude (Sooknanan and Malek, 1995). However, the greatest benefit is that NASBA can selectively amplify RNA even in the presence of genomic DNA (Greijer et al., 2000; Voisset et al., 2000). Despite these merits, NASBA is not truly isothermal, as denaturation steps are required, and reaction enzymes must be added separately because of their thermolability. Additionally, only short nucleotides in the range of 120–250 base pairs (bp) can be efficiently amplified (Katherine Loens et al., 2005).

NASBA has been widely applied and developed since first described in 1991. Multiplex NASBA was demonstrated as early as 1999 (van Deursen et al., 1999). Duplex NASBA (Jean et al., 2002; Klerks et al., 2001; van Deursen et al., 1999) has been combined with diverse detection methods (as mentioned above), whereas real-time analysis has become the most coupled detection method as multiplex capabilities have increased (Loens et al., 2008). However, the sensitivity of multiplex NASBA is decreased compared to the equivalent single-plex assay. Commercial kits that integrate NASBA with sample extraction or/and detection have been applied in clinical diagnosis. One example is the CorisBio-concept (Gemblux, Belgium), which incorporates NASBA with a lateral flow device (*Leishmania* OligoC-Test) for the diagnosis of *Leishmania* species (Bioconcept, 2012). Another is the NucliSENS EasyQ System (NucliSens® HIV-1 QT) developed by BioMeriexn (Marcy l'Etoile, France) that combines the sample extraction, amplification, and detection for quantification of the HIV-1 RNA in human plasma (Ginocchio et al., 2003). Miniaturisation of NASBA onto a lab-on-a-chip system in both nano and microliter volumes has been accomplished with real-time NASBA technology using molecular beacon detection. Sensitivities of the chip versions were comparable to the laboratory-based NASBA, yet with faster reaction times, lower costs and reduced contamination (Dimov et al., 2008; Gulliksen et al., 2004).

3.1.2. Signal-mediated amplification of RNA technology

An alternative RNA transcription-based isothermal amplification method is signal-mediated amplification of RNA technology (SMART). Unlike NASBA (TMA and 3SR), where the target sequences are directly amplified, SMART generates a target-dependent signal that can be further amplified (Wharam et al., 2001). Both DNA and RNA targets can be detected by this method. The system utilises two single-stranded oligonucleotide probes, an extension probe and a template probe. The extension probe is shorter in length than the template probe, but has a free 3' hydroxide for template extension. The template probe consists of a single-stranded T7 RNA polymerase promoter sequence (non-functional) and sequences to allow the capture and detection of the RNA signal (Fig. 4B) (Wharam et al., 2001). Both probes contain one region that can hybridise to the target sequence and a second much shorter region (6–8 nucleotides) that hybridises between

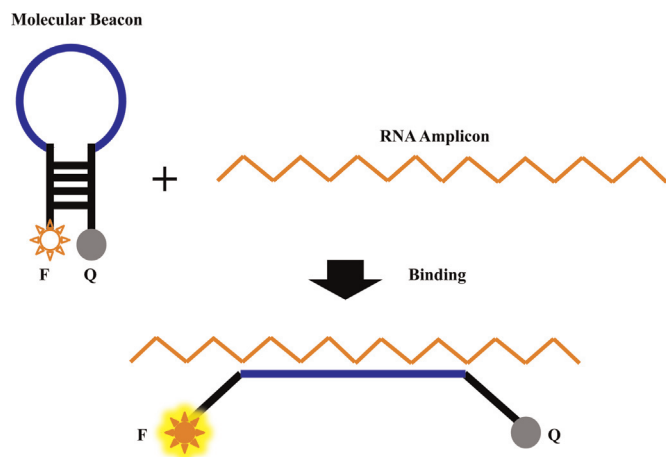


Fig. 3. Molecular beacon detection. The fluorophore (F) and quencher (Q) are brought in close proximity due to base pairing. Q absorbs all the energy from F, preventing detection of fluorescence. Upon binding of the complementary sequence to the nucleic acid target, the F and Q are separated from each other, allowing emission of fluorescence and subsequent detection.

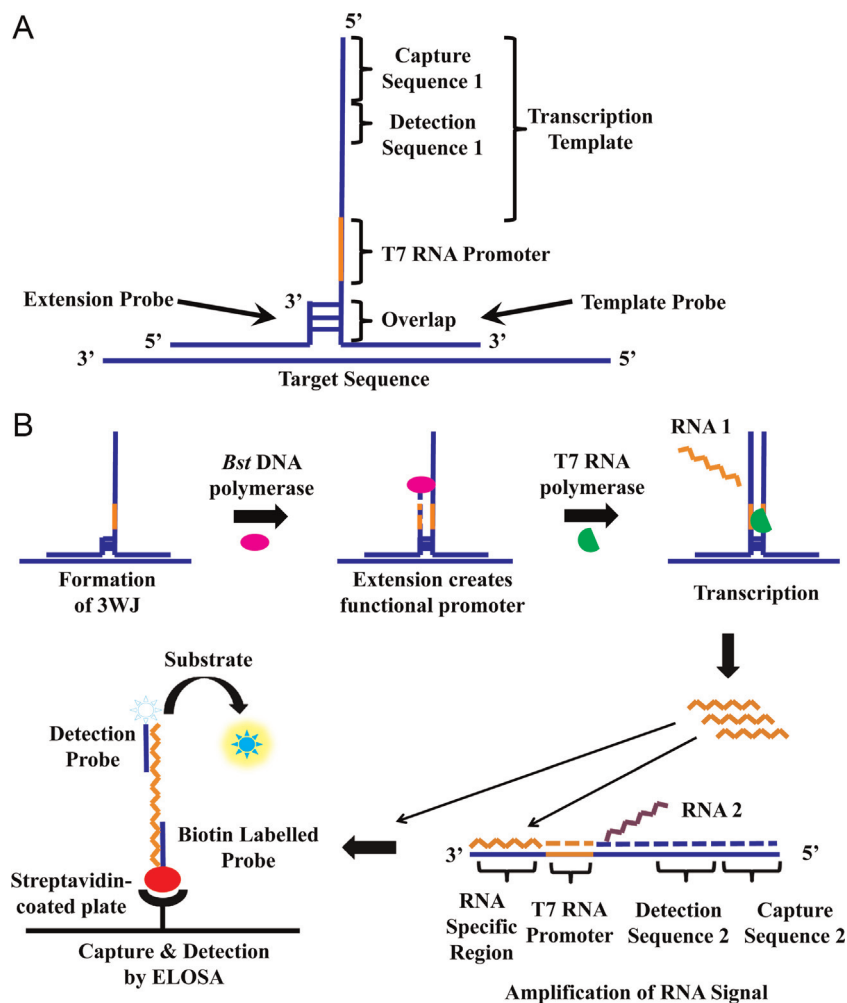


Fig. 4. SMART mechanism. (A) Structure of three-way junction (3WJ). A denaturation (at 95 °C) is performed at the beginning of the reaction, followed by addition of the two enzymes and isothermal amplification at 41 °C. Formation of the 3WJ allows the *Bst* DNA polymerase to extend the template from the 3' hydroxide end of the extension probe, which in turn generates a double-stranded T7 RNA polymerase promoter (functional) from which transcription of multiple copies of an RNA signal (RNA 1) occurs via T7 RNA polymerase. (B) The SMART assay. The resulting RNA 1 can be detected directly by ELOSA, or can anneal to a third probe (RNA amplification probe) to produce a second signal (RNA 2) that undergoes the same transcription mechanism as the first signal generation process.

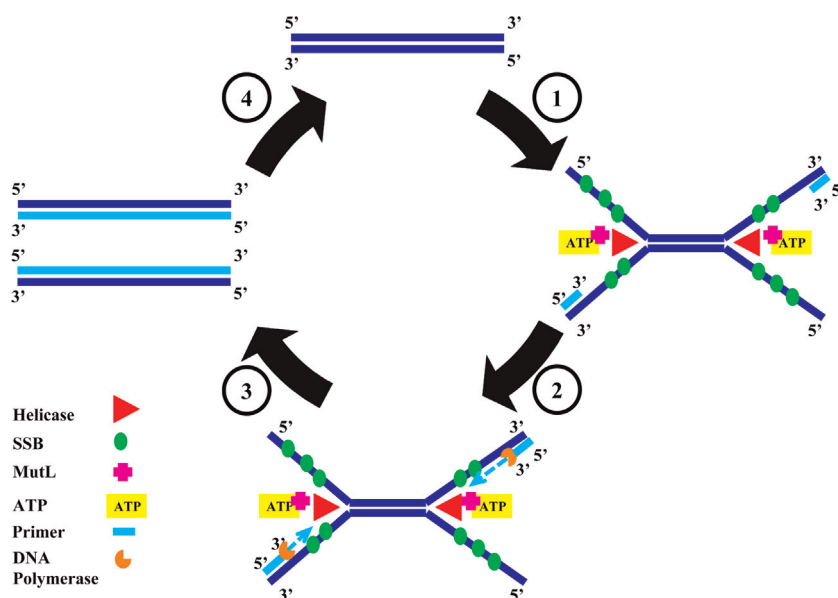


Fig. 5. HDA amplification scheme. DNA is separated by the enzyme helicase with the help of its co-factors—methyl-directed mismatch repair protein (MutL) and adenosine triphosphate (ATP). The separated strands are stabilised by single strand binding proteins (SSB), allowing primers to annealing followed by elongation to form new strands. Two duplex DNAs are subsequently formed and the process recycles.

the two probes (Wharam et al., 2001). Formation of a classic three-way junction (3WJ) structure (Fig. 4A) between the two probes and the target sequence provides high specificity of RNA signal amplification, as the two probes can only anneal to each other in the presence of the specific target. The entire process requires two enzymes, *Bacillus stearothermophilus* (*Bst*) DNA polymerase and T7 RNA polymerase, which function together under the same reaction conditions.

Detection of the generated RNA signals can be achieved by enzyme-linked oligosorbent assay (ELOSA). The RNA signals hybridise to an oligonucleotide linked enzyme molecule and a biotin-labelled oligonucleotide via specific sequences. This complex is then captured by streptavidin-coated wells through a biotin-labelled oligonucleotide, while detection is visualised by colour change of the solution once the enzyme substrate is applied (Fig. 4B).

Due to the unique formation of the 3WJ, SMART is a very specific amplification method. While the number of probes involved seemingly complicates the probe design, each set of the probes can be repeatedly used for detection of different targets, and only the target hybridising regions need to be redesigned each time. The sensitivity of SMART is less than that of PCR, yet SMART is tolerable to various samples, such as genomic DNA, total RNA, crude cultures and presence of non-target genomic DNA (Hall et al., 2002; Wharam et al., 2001). Future development for SMART aims to improve the sensitivity, to develop more efficient detection methods other than ELOSA and to couple amplification and detection in a single tube.

3.2. Methods based on DNA replication with enzymatic duplex melting/primer annealing

An alternative to using heat to separate the target nucleic acid, as described in the preceding section, another class of isothermal amplification protocols use enzymes to execute this step. This further models *in vivo* nucleic acid replication, simplifies the reaction mechanism, and also makes replication truly isothermal.

3.2.1. Helicase-dependent amplification

Helicase-dependent amplification (HDA) mimics the replication fork scheme created during DNA replication *in vivo*. Instead of using heat (as with NASBA and SMART above), helicase is used to separate the double-stranded DNA (Vincent et al., 2004). In the first report of HDA, Vincent et al. (2004) utilised *Escherichia coli* (*E. coli*) UvrD helicase (Runyon and Lohman, 1989) plus its accessory protein methyl-directed mismatch repair protein (MutL) (Mechanic et al., 2000), and cofactor adenosine triphosphate (ATP) to unwind the double-stranded DNA. Single strand binding proteins (SSB), either bacteriophage T4 32 proteins (Casas-Finet and Karpel, 1993) or RB 49 gene 32 proteins (Desplats et al., 2002), prevented the unwound DNA from re-annealing and enabling the two primers to bind. An *exo*–Klenow fragment of DNA polymerase I then extended the template from the two primers, thus producing another copy of the double-stranded template and the cycle was repeated continuously (Fig. 5). These results demonstrated that the HDA was successfully performed at 37 °C and would amplify genomic DNA from human blood samples (Vincent et al., 2004). Limitations included low speed and processivity of the *E. coli* UvrD helicase, and the lack of coordination between the helicase and the DNA polymerase (Wharam et al., 2001). Fast detection methods, such as real-time (Ramalingam et al., 2009) or lateral flow devices (Chow et al., 2008) are usually coupled to HDA.

An improvement of HDA was reported by An et al. (2005), who extended the mesophilic system mentioned above to a thermophilic HDA (tHDA) system by using thermostable UvrD helicase and MutL homologue from *Thermoanaerobacter tengcongensis*

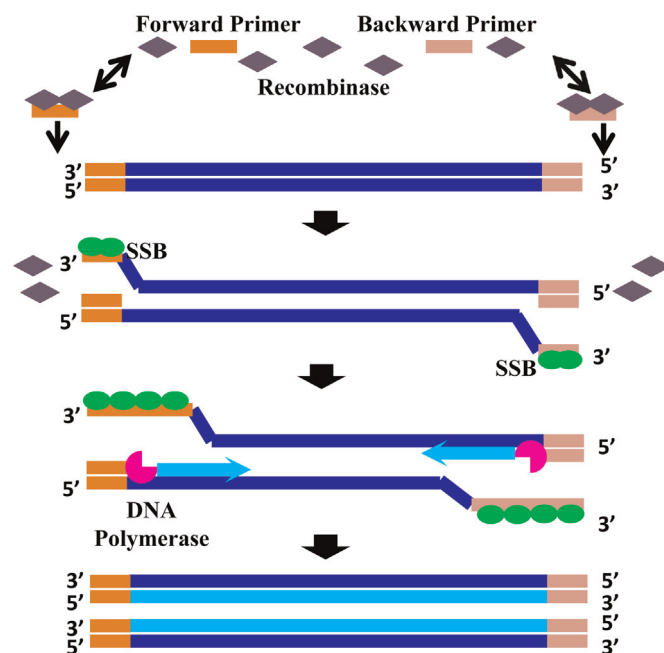


Fig. 6. RPA amplification scheme. RPA starts with the binding of recombinase (T4 uvsX) to forward and reverse primers, and stimulates the resulting complex to search for the homologous sequences in duplex DNA. Once the homology is located, a strand exchange reaction is performed. Single strand binding proteins (SSB, T4 gp32) bind to the parental strands, preventing them from interacting with the displaced template strands. DNA polymerase (the large fragment of *Bacillus subtilis* Pol I, Bsu) then initiates template synthesis from the free 3' hydroxide of the two primers, forming two duplexes of DNA. Repetition of the cycle leads to exponential amplification.

(*Tte*). Their results showed that the *Tte*-UvrD helicase no longer relied on MutL and SSB as opposed to the observation in the mesophilic system (An et al., 2005). The *Tte*-UvrD helicase also allowed the amplification to be performed at higher temperature (60–65 °C) with improved sensitivity and specificity (An et al., 2005). Furthermore, the *Tte*-UvrD helicase was found to be the best partner for *Bst* DNA polymerase in tHDA (An et al., 2005). Another improvement was made by Motre et al. (2008) who generated a new enzyme, named helimase by fusing the *Tte*-UvrD helicase with the *Bst* DNA polymerase to tackle the inefficient coordination of the helicase and the DNA polymerase. The helimase showed a much higher processivity in comparison to the regular HDA system, which led to an increase of amplification length, from 70 to 120 bp of the original HDA system to a maximum of 2.3 kilobases (Motre et al., 2008).

Goldmeyer et al. (2007) further developed the tHDA system for diagnostic applications. They demonstrated the first reverse transcription-tHDA (RT-tHDA) of different RNAs with real-time detection (Goldmeyer et al., 2007). In the following year, they reported that the HDA is suitable for rapid detection of *mecA* and *nuc* genes of methicillin-resistant *Staphylococcus aureus* (MRSA) (Goldmeyer et al., 2008). Another advancement was the combination of HDA with microarray technology. Andresen et al. (2009) successfully showed OnChip-HDA detection of MRSA and *Neisseria gonorrhoeae* (NG) individually and simultaneously. The detection sensitivities for simplex MRSA and NG were as low as 250 pg and 1 ng of genomic DNAs, respectively. While the duplex detection of both pathogens were ten times less sensitive than the simplex detections, the sensitivity of this OnChip-DNA was higher than the solution based HDAs previously described by An et al. (2005) and Goldmeyer et al. (2008).

Due to the simple reaction scheme, low incubation temperature and no requirement for denaturation, HDA is a promising

candidate for POC diagnosis. Further improvements could include adapting HDA to lab-on-chip systems that incorporate multistep sample processing and multiplex detection ability, to realise spatial reduction and automation.

3.2.2. Recombinase polymerase amplification

Piepenburg et al. (2006) revolutionised the previously described HDA system, by introducing an isothermal amplification method called recombinase polymerase amplification (RPA). Similar to HDA, RPA employs an enzyme to separate double-stranded DNA rather than by heating it (Fig. 6). RPA is very efficient, as billions of DNA copies can be generated within 40–60 min at incubation temperatures between 37 °C and 42 °C.

Methods that are used to detect PCR or HDA products can be applied to detect RPA products. However, for lateral flow and real-time detection, RPA has unique probes (corresponding to TwistAmp™ LF probe (Limited, 2013b) and TwistAmp™ exo probe (Limited, 2013a), respectively) to improve amplification efficiency. The TwistAmp™ LF probe (Fig. 7A) contains a fluorophore at its 5' end and a block at its 3' end, between which is embedded a tetrahydrofuran (THF) abasic-site mimic. Cutting by *E. coli*

endonuclease IV (Nfo) at the abasic-site releases the block, allowing incorporation of the probe into the amplicon. The opposing primer is also labelled, resulting in dual-labelled amplicons that can be detected via a lateral flow sandwich assay. The structure of the TwistAmp™ exo (Fig. 7B) probe is similar to the TwistAmp™ LF probe, except for the position of the fluorophore label, and an introduction of a quencher behind the abasic-site. This time, cutting by Exonuclease III (exo) separates the fluorophore and the quencher, allowing real-time detection. Applying these probes largely decreases the non-specific signals in the amplification, improving the specificity and sensitivity of the assay. However, both primers (30–35 bp) and the probe (46–52 bp) are quite long compared to the other methods (e.g. PCR and HDA etc.), and thus RPA is best suited to long template amplification.

RPA is a relatively new method that is yet to be widely adopted. Nevertheless, its simple reaction scheme, low reaction temperature, high efficiency, high specificity, and low sensitivity, will all contribute to the procedure likely becoming a favoured method for POC diagnosis. Rohrman and Richards-Kortum (2012) employed RPA coupled with lateral flow dipstick detection of HIV DNA. The results successfully demonstrated detection of HIV DNA (10 copies before amplification) in 15 min. Abd El Wahed et al. (2013) developed a reverse transcription RPA assay for the real-time detection of Foot-and-Mouth virus within 10 min. This assay was later applied to in-field detection (using a hand-held real-time scanner) during the Foot-and-Mouth disease outbreak in Egypt in 2013. The clinical sensitivity (98%) was almost equal to the reverse transcription PCR detection (100% sensitivity). Additionally, RPA has been adopted into a lab-on-a-chip system. Lutz et al. (2010) miniaturised RPA into a microfluidic system coupled with real-time detection. The system consisted of several reaction chambers fused onto a foil-based centrifuge that is equipped with an incubator. The pre-stored reaction buffer and dry reagents in different chambers were flowed into a reaction chamber and mixed via centrifugation. This automation system allowed detection of 20 reactions (10 µL per reaction volume) in parallel in less than 20 min with extremely high sensitivity (< 10 DNA copies of methicillin-resistant *S. aureus*). In the following year, Shen et al. (2011) expanded the detection throughput to 1650 parallel reactions and reduced the reaction volume to nano liters (9 nL) on a digital slip chip. The sensitivity was as low as 1:10⁵ dilution of 5 ng/µL of methicillin-resistant *S. aureus* genomic DNA. Moreover,

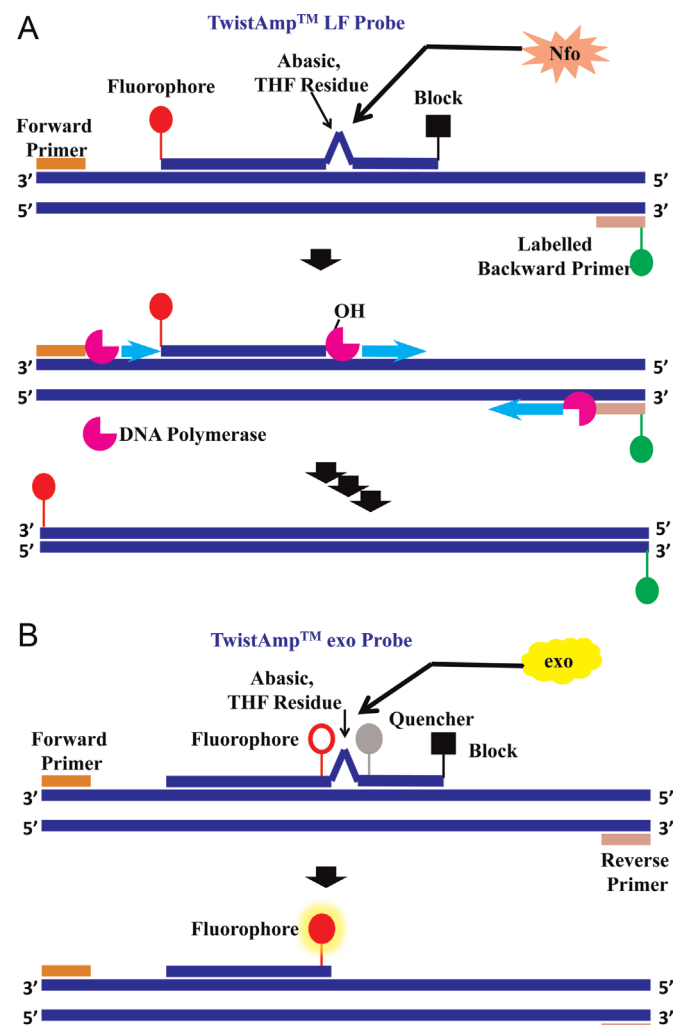


Fig. 7. RPA probes for lateral flow and real-time detections. (A) The TwistAmp™ LF probe for the lateral flow detection. Probe bound to DNA template contains a 3' block to prevent extension. The tetrahydrofuran (THF) is recognised and cleaved by *Escherichia coli* endonuclease IV (Nfo), which releases the block and allows incorporation into the amplicon via Bsu polymerase, which elongates from the 3' end hydroxide. (B) The TwistAmp™ exo probe for the real-time detection. Fluorescent signals are generated by separating the fluorophore and the quencher via Exonuclease III (exo) cutting at the abasic-site.

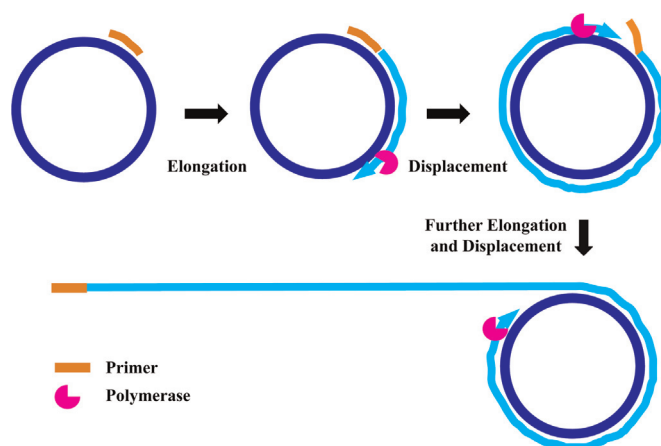


Fig. 8. RCA reaction scheme. The reaction is initiated by a single primer binding onto the circular single-stranded DNA template. The Phi29 bacteriophage DNA polymerase (φ29DNAP) then elongates around the template and eventually completes the circle. The DNA amplification is continuous, as the newly synthesised strand keeps on replacing the previously generated strand under the strand displacement activity of the φ29DNAP. The amplification can go on indefinitely until certain reactants are depleted.

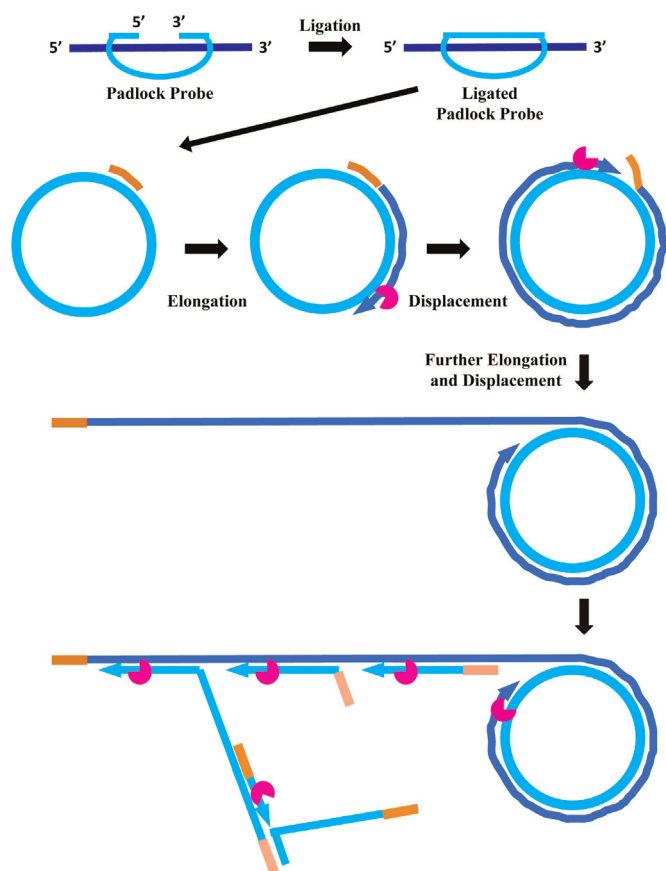


Fig. 9. RAM reaction scheme. The padlock probe incorporates two complementary regions to the target DNA at its 5' and 3' ends, respectively. Binding of the probe onto the target DNA brings the two ends into juxtaposition, enabling them to be covalently joined by DNA ligase (Nilsson et al., 1994; Zhang et al., 1998). Joining these two ends connects the probe to the target DNA through catenation. Consequently, forward primers anneal to the linker region of the ligated padlock probe, and amplification occurs in an RCA manner, which produces multimeric single-stranded DNAs. The reverse primers then bind to the nascent single-stranded DNAs and are elongated by ϕ 29DNAP. Strand displacements take place when the extensions reach the downstream-bound reverse primers with their extended sequences. These displaced DNA strands subsequently serve as templates for further primer extension and amplification.

Hakenberg et al. (2012) used RPA to successfully demonstrate the first passive laminar flow mixing of reagents on a microfluidic system. This system enabled efficient solvent mixing while reducing space on a chip. Finally, the RPA reaction reagents are available as commercial kits with freeze-dried reagents. All these developments indicate that RPA is rapidly gaining popularity as a POC diagnostic, and further advances to achieve fully automated diagnostic devices are likely to appear in the near future.

3.3. Methods based on DNA-polymerase-mediated strand displacement from linear/circular targets

Beyond the use of DNA enzymes to open replication forks for isothermal amplification, other protocols use circular nucleic acids or involve circular nucleic acid structures during amplification. These methods use DNA polymerases that have strand displacement abilities to generate continuous growing nucleic acid in a self-displacing cycle.

3.3.1. Rolling circle amplification

DNA replication not only uses linear DNA, but can also employ circular DNA as a template for DNA replication. *In vivo*, replication

of plasmids or viral nucleic acids can occur *via* a rolling circle mechanism (Kornberg and Baker, 1992). *In vitro*, this refers to the rolling circle amplification (RCA) (Fire and Xu, 1995; Liu et al., 1996). RCA relies on the high polymerase processivity and strand displacement activity of ϕ 29 bacteriophage DNA polymerase (ϕ 29DNAP) to synthesise tandem repeats of DNAs (Blanco et al., 1989; Fire and Xu, 1995; Liu et al., 1996). The resulting amplicons are the multiple repeats of the circular template (Fig. 8), which can subsequently be digested by enzymes into each individual short oligonucleotides (Liu et al., 1996).

The benefits of RCA are two-fold: the reaction mechanism is simple, only requiring a single primer and a productive enzyme; and the reaction is isothermal, with an incubation temperature as low as 23 °C. The circular DNA is the predominant template for amplification even in the presence of linear DNA, and the yield is very high. Such isothermal amplification also applies in RNA replication through the use of a circular RNA template (Daubendiek et al., 1995). The generation of RNA copies does not need an RNA promoter sequence, in contrast to the previously described NASBA and SMART methods. The end products of RCA can be visualised on a gel, or be monitored in real-time using molecular beacons (Nilsson et al., 2002), molecular zippers (Yi et al., 2006), or fluorogenic dyes (Di Giusto et al., 2005). RCA has a wide range of applications, from genomics (Schweitzer and Kingsmore, 2001) and proteomics (Kingsmore and Patel, 2003), to nanotechnology (Lin et al., 2006) and functional nucleic acids (Navani and Li, 2006).

While RCA is powerful, amplification occurs only linearly over time. To this end, several other isothermal amplification methods have been developed to enable exponential amplification of nucleic acids. One of these is the ramification amplification (RAM; Zhang et al., 2001a), hyper-branched RCA (Lizardi et al., 1998) or cascade RCA (Thomas et al., 1999). This reaction employs a padlock or C-probe (Nilsson et al., 1994) as a template for amplification (Fig. 9). The padlock probe consists of two complementary regions to the target DNA at its 5' and 3' ends, respectively. RAM differs from RCA by applying reverse primers, which bind to the nascent single-stranded DNAs and thus are elongated by the DNA polymerase ϕ 29DNAP (Zhang et al., 2001a). Strand displacements take place when the extensions reach the downstream-bound reverse primers with their extended sequences. These displaced DNA strands subsequently serve as templates for further primer extension and amplification. Such multiple primer extensions and amplifications progress simultaneously, thereafter generating ramified double-stranded DNAs exponentially (Zhang et al., 2001a, 2001b) (Fig. 9). By hooking the padlock probe onto the target DNA, RAM becomes ideal for *in situ* amplification, allowing the amplicons to be localised within cells for observation of their morphological characteristics.

Another derivative of RCA is multiple displacement amplification (Luthra and Medeiros, 2004; Spits et al., 2006b) (MDA), or multiply-primed RCA (Frank et al., 2001). Unlike RCA and RAM, MDA uses random hexamer primers to anneal to the circular DNA template, followed by extension through the polymerase. Displacement of each newly synthesised strand results from elongation of the primer behind it. Secondary priming events subsequently occur on the displaced strands, and a network of hyperbranched DNAs forms as more strands are replaced (Fig. 10) (Frank et al., 2001; Spits et al., 2006b). MDA allows amplification of circular or linear DNAs directly from cells or plaques, and yields 1–2 μ g of double-strand DNA from single cells within a few hours, which is even higher than that of RAM. Because of this, MDA is used in whole-genome amplification (Dean et al., 2002; Hosono et al., 2003) (WGA), BAC array-comparative genomic hybridisation (Le Caignec et al., 2006) (CGH), or short tandem repeat and mutation analysis (Spits et al., 2006a).

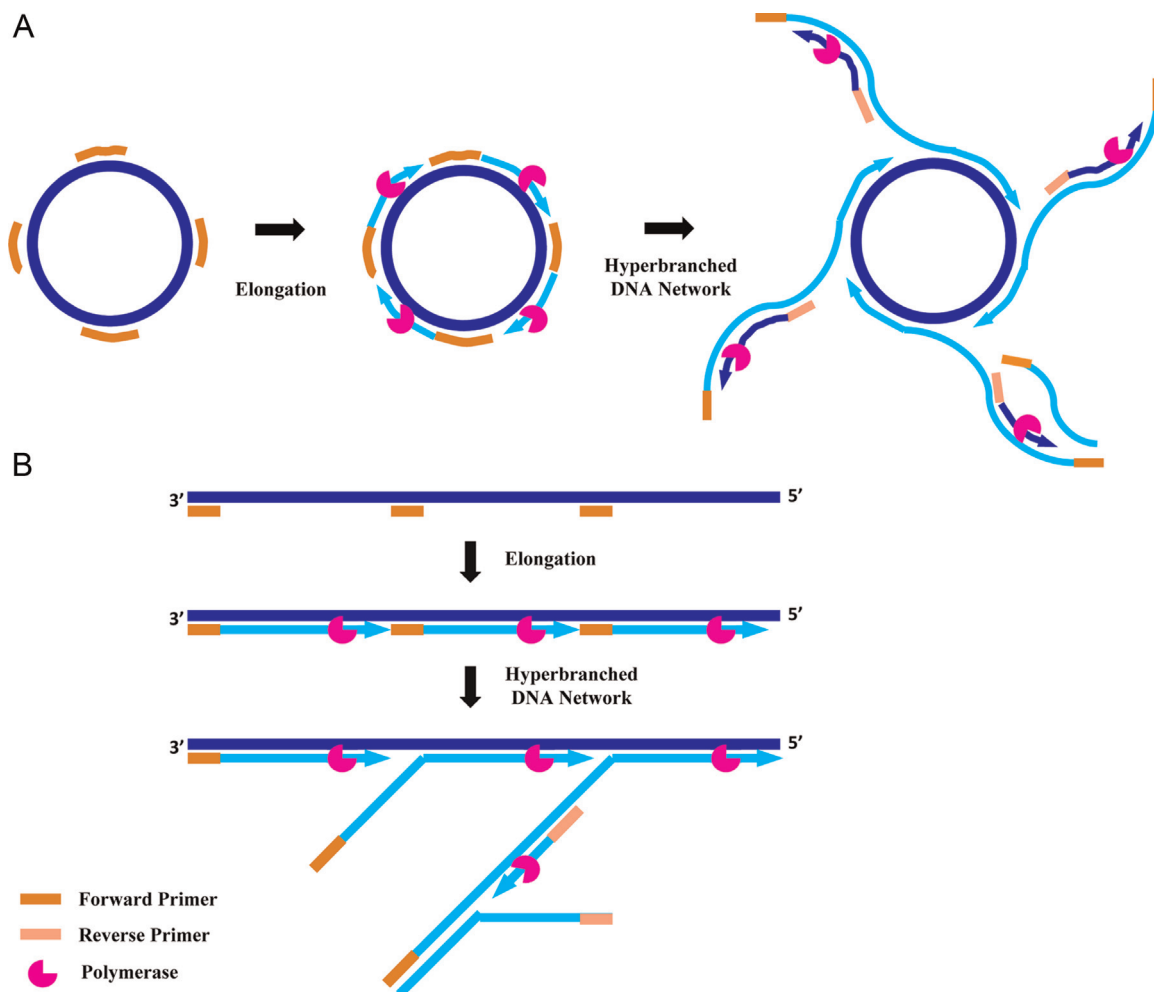


Fig. 10. MDA reaction scheme for (A) circular or (B) genomic DNA template. In both cases, hexamer primers anneal to the DNA template, and are extended by polymerase. Displacement of each newly synthesised strand occurs by elongation of subsequent primer extension occurring 5' to the existing extension area. Secondary priming events subsequently occur on the displaced strands, and a network of hyperbranched DNAs forms as more strands are replaced.

3.3.2. Loop-mediated isothermal amplification

Another method that utilises strand displacement and cyclic amplification is loop-mediated isothermal amplification (LAMP). This method employs a set of four primers to target six distinct regions on double-stranded DNA (Fig. 11A) (Notomi et al., 2000). The reaction progresses through two steps that are catalysed by *Bst* DNA polymerase with strand displacement activity: a non-cycling amplification step (Fig. 11B) and a cycling amplification step (Fig. 11C) (Notomi et al., 2000). The amplification is conducted at 65 °C for two reasons. Firstly, because the *Bst* polymerase has the best performance at this temperature, and secondly, because the double-stranded DNAs are at dynamic equilibrium around this temperature, so that one of the LAMP primers (FIP) can anneal to the DNAs without a denaturing step to initiate the synthesis (Nagamine et al., 2001; Notomi et al., 2000). During the non-cyclic amplification step, a series of primer binding and extension events enable the formation of a single-stranded double stem-loop DNA structure. This DNA structure then enables the reaction to enter into a cycling amplification cycle. The four primers operate in the same manner, but on growing amplified DNAs, thus leading to cauliflower-shaped products consisting of alternatively inverted repeats of the target sequence on the same strand. The LAMP amplification is efficient, as 10^9 copies of DNAs can be produced within an hour (Notomi et al., 2000). Uniform and single-stranded DNA from the LAMP products can also be obtained through enzyme digestion (Nagamine et al., 2002b).

Detection of LAMP products occurs either by end-point methods or by real-time assays. Agarose gel electrophoresis or lateral flow devices are used for end-point detection. Because of the LAMP mechanism, the amplicons appear as ladder-like bands on the gel. The lateral flow detection occurs via immunoassay using antibody-dependent or antibody-independent formats. Real-time LAMP product detection can be achieved through fluorescent dyes such as calcein (Tomita et al., 2008) or hydroxyl naphthol blue (Goto et al., 2009). Additionally, a unique feature of LAMP is that products can be visualised by the naked eye through the appearance of white turbidity formed by magnesium ions and pyrophosphates (Mori et al., 2001). Mori et al. (2004) developed a turbidimeter machine to monitor the turbidities of multiple samples simultaneously in real-time. This also allows quantification of the LAMP products. In addition, LAMP products can be detected in a sequence specific manner by adding low-molecular weight polyethylenimine (PEI) and fluorescently labelled probe (Mori et al., 2006). PEI cannot form an insoluble complex with a single-stranded anionic polymer with a low molecular weight (e.g. an oligo DNA probe), but it can do so with DNA with high molecular weight (e.g. the LAMP products) (Mori et al., 2006). The fluorescently labelled LAMP products are incorporated in the insoluble complex, thus different fluorescent colours distinguishes the sequences amplified.

LAMP was developed in 2000, and it remains the most widely applied isothermal amplification method. LAMP has extremely

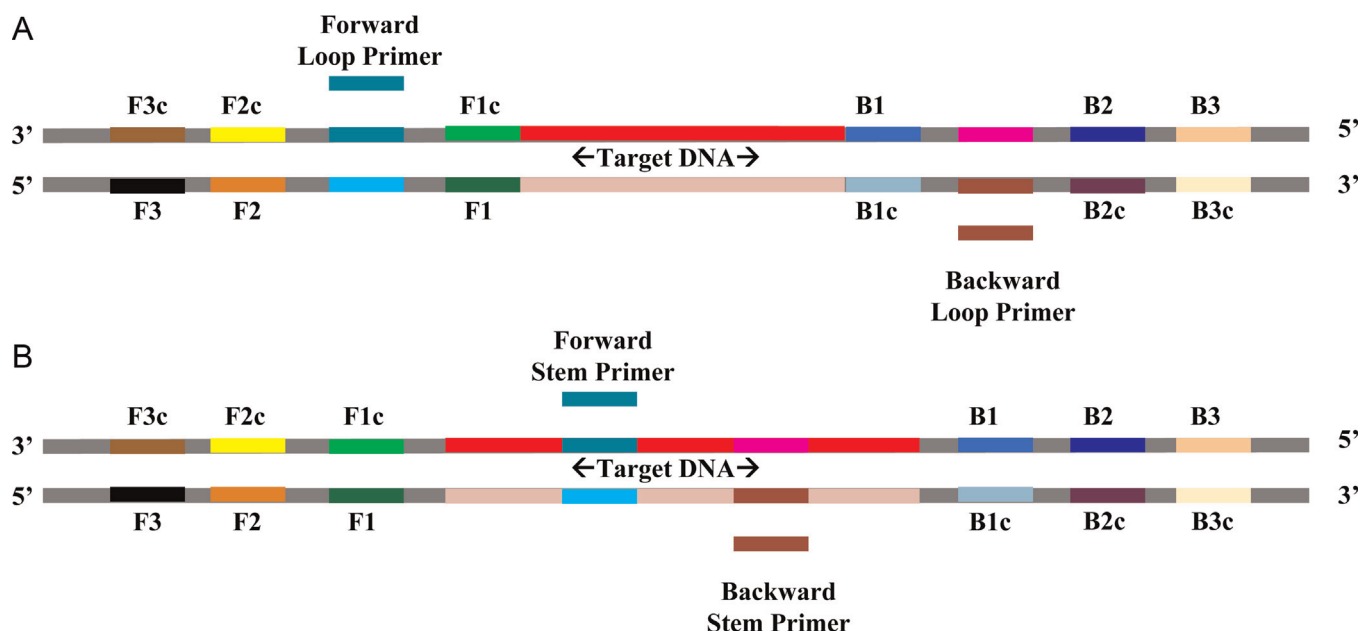


Fig. 12. Binding sites of extra primers on the target DNA of LAMP. (A) Location of loop primers of LAMP. (B) Location of stem primers of LAMP.

high specificity, as the reaction only occurs when the six distinct regions within the DNA target are recognised by the four primers. The LAMP reaction is very fast and efficient, and can be further accelerated by applying an extra pair of loop primers (Nagamine et al., 2002a) (Fig. 12A) or stem primers (Gandelman et al., 2011) (Fig. 12B). These extra primers bind to additional regions on the target nucleic acid, which increases the strand displacement activity on the formed stem loop structure during the cycling amplification step. LAMP can also tolerate inhibitory substances present in biological samples better than PCR (Kaneko et al., 2007). A series of reaction condition studies, ranging from amplification temperature, time, pH, sensitivity and linearity, carried out by Francois et al. (2011) demonstrated that LAMP is a robust reaction for diagnostic applications. Because of these characteristics, LAMP has been applied to pathogen detection (Fu et al., 2011; Mori and Notomi, 2009; Parida et al., 2008), including bacteria, viruses, fungi, parasites and mycoplasma, as well as genetically modified material, tumour detection, embryo sex identification, SNP analysis (Iwasaki et al., 2003; Li et al., 2011), discrimination of human gene copy number (Nakamura et al., 2010), and human RNA expression (Muto et al., 2011; Osako et al., 2011).

Improvements to the LAMP procedure being investigated include reducing electrical requirements of the assay for use in low-resource settings. LaBarre et al. (2011) developed an electricity-free heater based on exothermic chemical reactions and engineered phase change materials to enable incubation of a LAMP reaction, while Hatano et al. (2010, 2011) used pocket warmers as the heat source to drive the reaction. Another LAMP development is multiplexed reactions (Aonuma et al., 2010; He and Xu, 2011; Liang et al., 2012; Tanner et al., 2012), but so far improvements have only reached penta-plex detection due to the complexity of the LAMP primer design, and non-specific amplification products are sometimes a problem (Tanner et al., 2012). Multiplexing for high-throughput detection is also occurring, such as a microfluidic system of LAMP (Fang et al., 2010) and digital LAMP (Gansen et al., 2012; Zhu et al., 2012). These developments allow a high-level of detection efficiency (e.g. 1200 samples simultaneously) and sensitivity (e.g. 10 fg of DNA) but require only small sample volumes (e.g. 2 μ L) with a greater level of automation.

3.4. Methods based on polymerase extension/strand displacement and a single strand-cutting event

Isothermal amplification can also be performed using restriction enzyme nicking followed by polymerase displacement of the nicked strand, as a new strand is elongated from the nicking site. This is an alternative way of maintaining a self-cycling amplification step without the requirement for primers to initiate strand displacement (such as in the LAMP method above).

3.4.1. Strand displacement amplification

Strand displacement amplification (SDA) exploits (i) a restriction endonuclease to nick the unmodified strand of the DNA target and (ii) an exonuclease-deficient DNA polymerase [exo[−] Klenow polymerase (Derbyshire et al., 1988)] to displace the downstream DNA strand at the nick sites (Walker et al., 1992a, 1992b). The displaced strands then serve as templates for an antisense DNA reaction. Like NASBA, SDA requires a pre-heating step to denature the double-stranded DNAs, while the rest of the reaction is performed at 37 °C. The addition of the two enzymes occurs after the denaturing step. Similar to LAMP, SDA employs a set of four primers, two of them consist of complementary sequences to the target DNA at their 3' end and restriction enzyme sites of *HincII* at their 5' end. The other two are normal primers such as F3 and B3 in LAMP. The entire reaction is composed of two steps: a template generating step (Fig. 13A) and a cycling step (Fig. 13B), which occur in tandem to produce an exponential amount of double-stranded DNAs. The amplicons can be detected by gel electrophoresis, or hybridisation based technology (Chen et al., 2009; Little et al., 1999).

SDA is the precursor of MDA and LAMP. It was developed to provide an alternative DNA amplification method for POC testing in the clinic. The breakthrough for SDA adoption in clinical systems came with the advent of real-time homogeneous detection chemistry. The BD ProbeTec™ ET System (Company, 2014) (Becton Dickinson Microbiology Systems, Sparks, MD) is a real-time high-throughput SDA system that has been commercially available since 1999 for the diagnosis of *Chlamydia trachomatis* (CT), *N. gonorrhoeae* (NG), and Human Papilloma Virus (HPV) (Little et al., 1999). A multiplexed SDA has also been described (Walker et al., 1994), which simultaneously amplifies two target

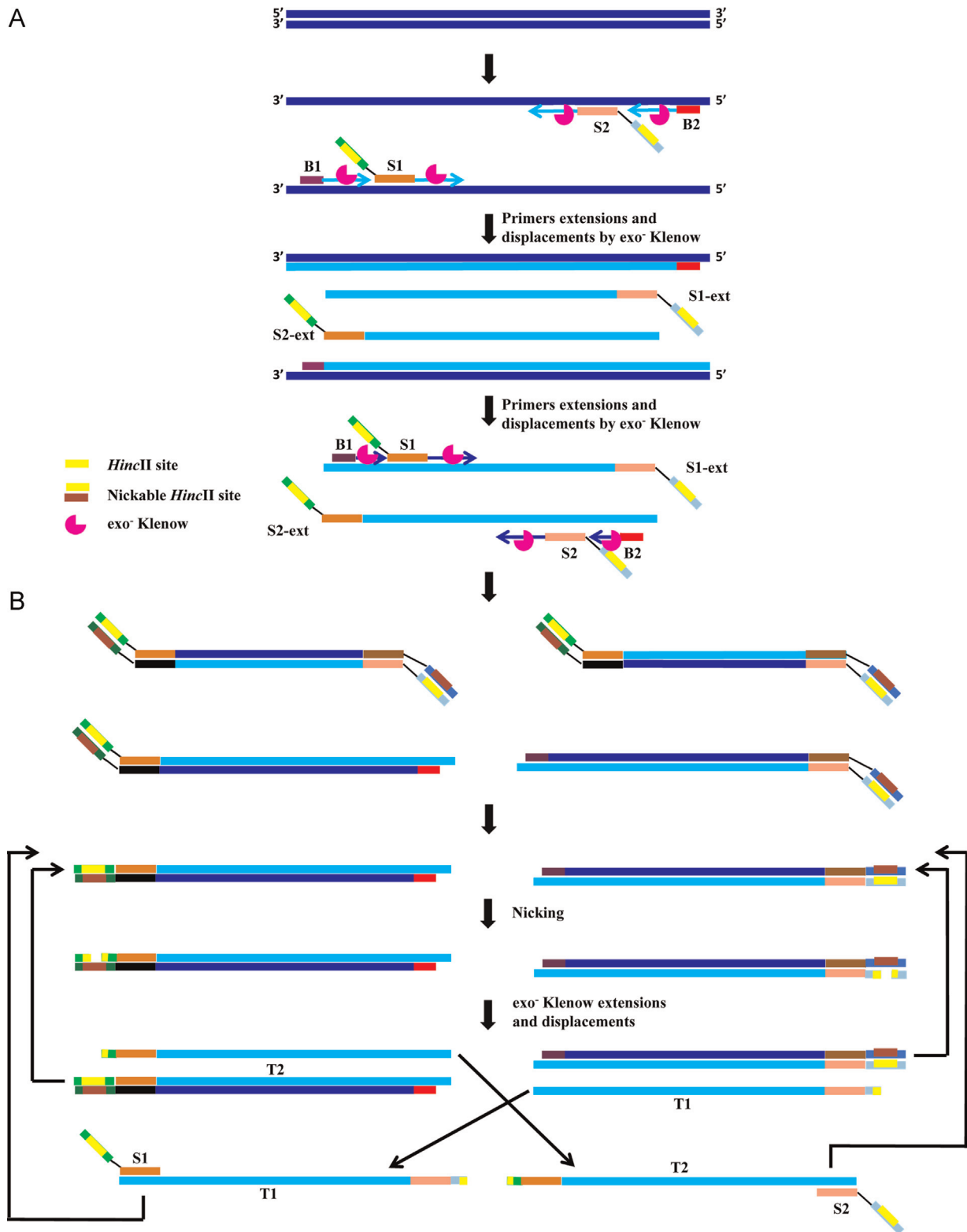


Fig. 13. SDA reaction scheme. (A) Template generating step. Once the DNA is denatured, the four primers (S1, S2, B1 and B2) bind their complementary sequences and are simultaneously extended by *exo- Klenow* DNA polymerase using dGTP, dCTP, dTTP and dATP(α S) (5'-O-L-thiotriphosphate that is resistant to the endonuclease activity). Extension of B1 displaces the S1 extension product (S1-ext) and extension of B2 displaces the S2 extension product (S2-ext). Subsequently, B2 and S2 bind to S1-ext, and B1 and S1 bind to S2-ext. Extensions and displacements then follow in the same manner. This results in two fragments with hemiphosphorothioate *HincII* at each end and another two fragments with a hemiphosphorothioate *HincII* site at just one end. (B) Cycling step of SDA reaction. The *HincII* enzyme nicking and DNA polymerase extension and displacement initiates upon the above four fragments. The displaced templates (T1 and T2) can be re-annealed by S1 and S2 respectively to become nickable fragments that are elongated by DNA polymerase, while the newly synthesised two double-stranded DNAs undergo another round of enzyme nicking, extension and displacement. *Note:* The two fragments with hemiphosphorothioate *HincII* at each end have a more complicated mechanism in the cycling step, thus only the two fragments with hemiphosphorothioate *HincII* at just one end were described in the example.

Table 1

A summary of isothermal amplification methods.

Method	Preferred amplicon	Aplicom type	Pre-heating	Reaction temperature (°C)	Primers	Enzymes	Efficiency	Key publication	Year
NASBA	RNA	RNA, DNA	Yes (65 °C for RNA and 95 °C for DNA)	41	2	3	10 ⁶ –10 ⁹ -Fold amplification within 90 min	Compton (1991)	1991
3SR	RNA	RNA, DNA	Yes	37	2	3	10 ⁹ -Fold amplification in less than 1 h	Guatelli et al. (1990)	1991
TMA	RNA and DNA	RNA, DNA	Yes	37	2	2	10 ⁶ -Fold amplification within 1 or 2 h	Gill and Ghaemi (2008)	1990
SMART	DNA	RNA, DNA	Yes (95 °C)	41	2	2	10 ⁴ –10 ⁵ -Fold amplification in 2 h	Wharam et al. (2001)	2001
HDA	DNA	RNA, DNA	No	37	2	2	10 ⁶ -Fold amplification in 2 h	Vincent et al. (2004)	2004
RPA	DNA	RNA, DNA	No	37–42	2	2	10 ⁴ -Fold amplification in 10 min	Piepenburg et al. (2006)	2006
RCA	Circular DNA	Multiple repeats of the circular template	No	23	1	1	9000 Nucleotides from a 34 nucleotides template within 1 h	Fire and Xu (1995)	1995
RMA	C-probe	Ramified double-stranded DNAs	No	35	2	2	> 8000 Nucleotides from the ligated C-probe within 1 h	Zhang et al. (2001a)	2001
MDA	Circular or linear DNAs	Ramified double-stranded DNAs	No	30	Random hexamer primers	1	10 ⁴ -Fold amplification in 8–10 h	Frank et al. (2001) , Spits et al. (2006b)	2006
LAMP	DNA	Various sized structures consisting of alternatively inverted repeats of the target sequence on the same strand	No	65	4 (6 or 8)	1	10 ⁹ -Fold amplification in less than 1 h	Notomi et al. (2000)	2000
SDA	DNA	DNA	Yes	37	4	2	10 ⁷ -Fold amplification in 2 h	Walker et al. (1992b)	1992

sequences and an internal control sequence in one test tube. Both targets were amplified 10^8 -fold within two hours, although the amplification efficiency was not as fast as individual reactions for each target sequence.

Although SDA is very efficient, focus moved to LAMP after its development, because LAMP is a true isothermal amplification and is more efficient than SDA. SDA can only amplify short target sequences ranging from 50 to 120 base pairs. Additionally, SDA was not very sensitive to DNA background, as non-specific products are not readily seen.

4. Emerging isothermal amplification methods

The proceeding sections covered isothermal amplification methods that are well established and have been widely adopted by the scientific community. Other isothermal technologies have also been described, which were developed as extensions of existing methods or as innovations. Examples include:

- (1) The exponential amplification reaction (EXPAR) is a mimic of the SDA method where releasing of DNA after nicking occurs due to the instability of the DNA duplex rather than the strand displacement activity of the enzyme. EXPAR aims to amplify short oligonucleotides (8–16 bp), and is used for characterising polymorphic sites in genomic DNA (Van Ness et al., 2003).
- (2) The single chimeric primer isothermal amplification (SPIA) (Kurn et al., 2005) was developed based on MDA and TMA methods. The amplicon generated from SPIA is single-stranded cDNA, which is useful for the transcriptome gene expression on-chip analysis.
- (3) The exonuclease III-induced cascade two-stage isothermal amplification-mediated zinc (II)-protoporphyrin IX/G-quadruplex supramolecular fluorescent nanotags (Xue et al., 2014), which apply the same strategy as SMART by generating a target-dependent signal that can be further amplified. This method reached a detection limit of 0.75 fM for 35 bp DNA, and successfully discriminated a single mismatch. The method holds great potential for early diagnosis in gene-related diseases.
- (4) The beacon assisted detection amplification (BAD AMP) (Connolly and Trau, 2010, 2011), which generates a target-dependent signal at the same time as the target itself is amplified. The BAD AMP aims to amplify and detect short DNA sequences (about 22 bp).
- (5) The hairpin fluorescence probe assisted isothermal amplification (Ma et al., 2014), which does the same as the BAD AMP method, and also simultaneously reversely transcribes the RNA into a more stable double-stranded DNA. This method aims to detect and amplify microRNA (an average of 22 nucleotides). Both this method and the BAD AMP have ultra-sensitivity that detects femto-molar (10^{-15}) concentrations of target.

Most of these new developing isothermal amplification methods are still undergoing improvements to achieve technological maturity. They may emerge as competitors to the more established methods described in the preceding sections, but have not yet been implemented beyond their original proof-of-concept description.

5. Conclusions and future perspectives

Here we reviewed isothermal amplifications that alleviate the usage of a thermal cycler in non-isothermal amplifications such as

PCR. Each method described has its own strengths and weaknesses (see Table 1) in terms of:

- (1) Reaction scheme: HDA, RPA and RCA have simple reaction schemes.
- (2) Primer design: HDA, RCA, RAM and MDA involve relatively easy primer design. Primer designed for PCR can be applied in these methods.
- (3) Amplification ability: HDA is capable of synthesising long targets; NASBA, 3SR, TMA, and SDA can only amplify short templates; RCA, RAM, and MDA generate long chain of tandem repeats.
- (4) Pre-heating requirement: HDA, RPA, RCA, RAM, MDA and LAMP do not require pre-heating of the template, thus are truly isothermal.
- (5) Application specificity: NASBA prefers to RNA template generation; RCA is circular DNA specific; RAM allows *in situ* amplification; MDA is capable of whole genome amplification.

As our knowledge of biological processes and pathways improves, more transformations from *in vivo* nucleic acids synthesis to *in vitro* setup are likely to appear. Moreover, our molecular engineering ability to manipulate enzymes, oligonucleotides, and their reaction networks is likely to lead to novel strategies that may be far removed from natural processes, such as the non-enzymatic amplification methods reviewed by Yan et al. (2014).

Due to the single temperature of reaction, isothermal amplification is a particularly promising candidate for low-resource settings application. Regardless of the underlying technology, key developments still needed for point-of-care application of isothermal amplification include: (1) balancing the requirements for improved speed, lower reaction volumes, reduced electricity requirements, and yet exhibiting high sensitivity and robustness; (2) employing higher multiplexed ability by allowing multiple (e.g. more than 5) reactions in one tube; and (3) being more amenable to be integrated with other processes, such as sample preparation, purification and detection, which paves the way for a fully automated system for POC application.

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